

Section 8.1: Introduction

Goal:

Some examples:

1. Estimate the proportion, p , of Americans who approve of the job President Trump is doing.
2. Estimate the average cost, μ , of a gallon of gas.
3. Estimate the standard deviation, σ , of the width of a car door.

Def. The **target parameter** is

Two Types of Estimates:

1. Point estimate:

- Ex.: $\hat{p} = .4 \rightarrow$

2. Interval estimate:

- Ex.: $(.37, .43) \rightarrow$ We estimate the proportion of Americans who approve of the job Trump is doing to be _____.

Def. An **estimator** is

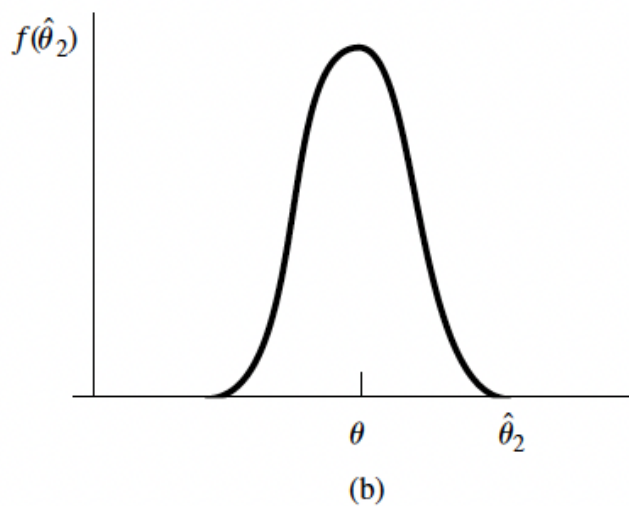
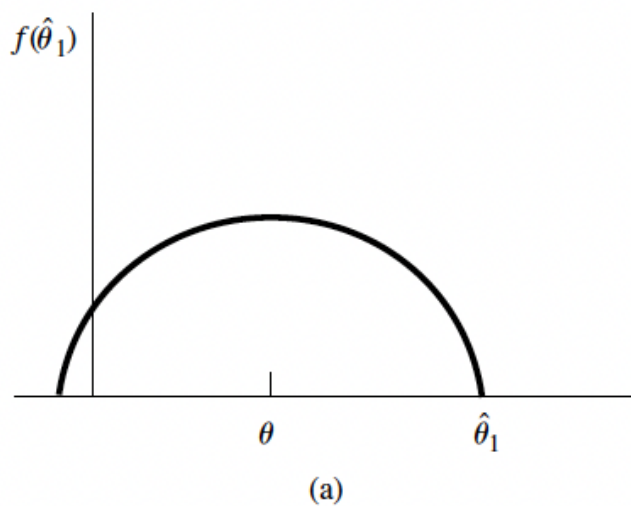
Section 8.2: The Bias and Mean Square Error of Point Estimators

Notation. In general, we will denote a parameter by _____ and an estimator by _____.

Def. $\hat{\theta}$ is an **unbiased estimator** of θ if

Def. The **bias** of a point estimator $\hat{\theta}$ is given by

Question. Suppose we have two unbiased estimators of θ , $\hat{\theta}_1$ and $\hat{\theta}_2$, with the following sampling distributions. Which is the better estimator?



Fact. Suppose $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ . Then, $\hat{\theta}_1$ is the better estimator if

Remark. One quantity we can use to measure the “goodness” of an estimator is the *mean square error (MSE)*.

Def.

Question. Which is better—a smaller or larger MSE?

Fact.

$$MSE(\hat{\theta}) =$$

Proof.

Example 1. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta = 3)$.

(a) Is $\bar{Y}$ an unbiased estimator of α ?

- (b) Find $MSE(\bar{Y})$.
- (c) Modify $\bar{Y}$ to get an unbiased estimator of α .
- (d) Find the MSE of the unbiased estimator from (c.).

Section 8.3: Some Common Unbiased Point Estimators

Remark. In Chapter 9, we will discuss methods for deriving “good” estimators. For now, we will consider some point estimators that make sense intuitively.

Def. The **standard error** of $\hat{\theta}$ is

Notation:

Table 8.1 Expected values and standard errors of some common point estimators

Target Parameter θ	Sample Size(s)	Point Estimator $\hat{\theta}$	$E(\hat{\theta})$	Standard Error $\sigma_{\hat{\theta}}$
μ	n	$\bar{Y}$	μ	$\frac{\sigma}{\sqrt{n}}$
p	n	$\hat{p} = \frac{Y}{n}$	p	$\sqrt{\frac{pq}{n}}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{Y}_1 - \bar{Y}_2$	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}^{*\dagger}$
$p_1 - p_2$	n_1 and n_2	$\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$	$\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}^{\dagger}$

* σ_1^2 and σ_2^2 are the variances of populations 1 and 2, respectively.

$\dagger$ The two samples are assumed to be independent.

Remark. In practice, we cannot always compute $\sigma_{\hat{\theta}}$, in which case we will estimate it! Examples where we must estimate $\sigma_{\hat{\theta}}$ are forthcoming.

Remark. As a result of the CLT, all point estimators in Table 8.1 are approximately Normal *if the sample sizes are large enough*.

Question. How large is “large enough?”

Rules of Thumb:

1. $\bar{Y}$:
2. $\hat{p}$:
3. $\bar{Y}_1 - \bar{Y}_2$:
4. $\hat{p}_1 - \hat{p}_2$:

Remark. Not all estimators are unbiased.

Example 2. Let $Y_1, \dots, Y_n$ be a random sample with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2$.

(a) Show that $S'^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$ is a biased estimator for σ^2 .

(b) Show that $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ is an unbiased estimator for σ^2 .

Section 8.4: Evaluating the Goodness of a Point Estimator

Def. The **error of estimation**, ϵ , is

Remark. In most practical situations, $2\sigma_{\hat{\theta}}$ is a good approximate bound for ϵ . Why?

Example 3. A study investigated the effects of a “Buckle Up Your Toddlers” campaign to get parents to use grocery cart seat belts. Investigators observed a random sample of parents in a large city and found that 192 of 594 of them buckled up their toddlers. Estimate p , the true proportion of parents who use grocery cart seat belts, and place a 2-standard-error bound on the error of estimation.

Section 8.5: Confidence Intervals

Def.:

- confidence interval (CI):
 - lower confidence limit:
 - upper confidence limit:
 - confidence coefficient:

Remarks.

- If $P(\hat{\theta}_L < \theta) = 1 - \alpha$, then $(\hat{\theta}_L, \infty)$ is a _____ one-sided CI.
- If $P(\theta < \hat{\theta}_U) = 1 - \alpha$, then $(-\infty, \hat{\theta}_U)$ is an _____ one-sided CI.

Pivotal Method: Find a pivotal quantity, U , that possesses two characteristics:

1.

2.

Using U , find a and b such that $P(a \leq U \leq b) = 1 - \alpha$.Finally, isolate θ in the middle.**Example 4.** Suppose we are to obtain a single observation Y from an exponential distribution with mean θ . Find a 90% confidence interval for θ .

Example 4 cont'd. Suppose we take a random sample of size $n = 1$ and observe $Y = 3$. Determine the 90% confidence interval for θ .

Example 5. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\theta)$. Find a 90% confidence interval for θ .

Recall: Suppose $Y \sim \text{Gamma}(\alpha = \frac{n}{2}, \beta)$, where n is a positive integer. Let $W = \frac{2Y}{\beta}$. Then,

$$W \sim \chi_n^2.$$

Example 5 cont'd. Suppose we take a random sample of size $n = 3$ and observe $\bar{Y} = 3$. Determine the 90% confidence interval for θ .

Example 6. Suppose we take a random sample of size $n = 1$ from a $U(0, \theta)$ distribution. Find a 95% lower confidence bound for θ .

Section 8.6: Large-Sample Confidence Intervals

We will now discuss how to find CIs for the target parameters in Table 8.1.

Table 8.1 Expected values and standard errors of some common point estimators

Target Parameter θ	Sample Size(s)	Point Estimator $\hat{\theta}$	$E(\hat{\theta})$	Standard Error $\sigma_{\hat{\theta}}$
μ	n	$\bar{Y}$	μ	$\frac{\sigma}{\sqrt{n}}$
p	n	$\hat{p} = \frac{Y}{n}$	p	$\sqrt{\frac{pq}{n}}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{Y}_1 - \bar{Y}_2$	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}^{*\dagger}$
$p_1 - p_2$	n_1 and n_2	$\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$	$\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}^{\dagger}$

* σ_1^2 and σ_2^2 are the variances of populations 1 and 2, respectively.

$\dagger$ The two samples are assumed to be independent.

Recall. As a result of the CLT, all point estimators in Table 8.1 are approximately Normal *if the sample sizes are large enough*.

- This implies $Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$ is approximately _____.

Question. Is Z a pivotal quantity? Why or why not?

Example 7. Suppose $\hat{\theta} \dot{\sim} N(\theta, \sigma_{\hat{\theta}})$. Using the Pivotal Method, find a $100(1 - \alpha)\%$ CI for θ .

Facts: If $\hat{\theta} \sim N(\theta, \sigma_{\hat{\theta}})$,

- a $100(1 - \alpha)\%$ CI for θ is $(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}}, \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}})$.
- a $100(1 - \alpha)\%$ lower one-sided CI for θ is _____.
- a $100(1 - \alpha)\%$ upper one-sided CI for θ is _____.

Example 8. Find a 95% confidence interval for μ if $n \geq 30$.

Example 9. A study investigated the effects of a “Buckle Up Your Toddlers” campaign to get parents to use grocery cart seat belts. Investigators observed a random sample of parents in a large city and found that 192 of 594 of them buckled up their toddlers.

(a) Find a 90% confidence interval for the true proportion of parents who use grocery cart seat belts.

(b) Interpret this interval in context of the problem.

Example 10. In a study of women science majors, the following data were obtained on two groups, those who left their profession within a few months after graduation (leavers) and those who remained in their profession after they graduated (stayers).

	Leavers	Stayers
Sample Size	103	225
Mean GPA	3.16	3.28
Standard Deviation	0.52	0.46

- (a) Find a 99% confidence interval for the difference in mean GPA of the two groups.
- (b) Based on your answer to (a), is there significant evidence that the mean GPA for the leavers differs from the mean GPA for the stayers? Why or why not?
- (c) Interpret this interval in context of the problem.

Question. What does it mean to be 99% confident?

Section 8.7: Selecting the Sample Size

Example 11. Refer to the “Buckle Up Your Toddlers” example. Suppose we want to estimate the true proportion of parents who use grocery cart seat belts, and we want the error of estimation to be less than .005 with probability 0.90. What sample size do we need to obtain an error of estimation that is this small? Suppose we know from previous studies that $p = 0.15$.

Example 11 cont’d. If we had no prior knowledge of what p is, what sample size should we use?

Section 8.8.1: Small-Sample Confidence Intervals for μ

Motivation: We want to find a $100(1 - \alpha)\%$ CI for μ , but σ is unknown and $n < 30$.

- Problems:

1. sampling distribution of $\bar{Y}$ may not be Normal
2. s may not be a good estimate of σ

To solve these problems, we add the following assumption about our data:

Assumption:

Example 12. Using the Pivotal Method, find a $100(1 - \alpha)\%$ CI for μ .

Recall. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Then, $T \sim t_{n-1}$.

Facts: Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then,

- a $100(1 - \alpha)\%$ CI for μ is $\left(\bar{y} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{y} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}\right)$.
- a $100(1 - \alpha)\%$ lower one-sided CI for μ is _____.
- a $100(1 - \alpha)\%$ upper one-sided CI for μ is _____.

Note: These CIs are appropriate for samples of *any* size—not just when $n < 30$.

Example 13. An organic chemist randomly sampled ten 4.85-g aniline specimens to produce acetanilide. The dry yield of acetanilide was measured for each specimen—the mean dry yield was 3.781 grams, and the standard deviation was 0.181 grams. Prior research shows that dry yields are normally distributed.

(a) Construct a 95% confidence interval for the true mean dry yield of acetanilide.

(b) Interpret this interval in context of the problem.

Section 8.8.2: Small-Sample Confidence Intervals for $\mu_1 - \mu_2$

Assumption: We have two independent random samples: $X_1, \dots, X_{n_1} \stackrel{i.i.d.}{\sim} N(\mu_1, \sigma_1)$ and $Y_1, \dots, Y_{n_2} \stackrel{i.i.d.}{\sim} N(\mu_2, \sigma_2)$.

Motivation: We want to find a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$, but σ_1 and σ_2 are unknown, and either $n_1 < 30$ or $n_2 < 30$ (or both).

Remark. The CI formula depends on whether $\sigma_1^2 = \sigma_2^2$ is a reasonable assumption.

- Regardless, the forthcoming formulas are appropriate for samples of *any* size.

Notation. The quantities s_1 and s_2 represent the sample standard deviations of groups 1 and 2, respectively.

Case 1 ($\sigma_1^2 = \sigma_2^2$): Use the pooled procedure.

CI Formula:

Case 2 (σ_1^2 and σ_2^2 are not necessarily equal): Use the non-pooled procedure.

CI Formula:

$$df = \frac{[(s_1^2/n_1) + (s_2^2/n_2)]^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

Section 8.9: Confidence Intervals for σ^2

Motivation: Suppose $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Find a CI for σ^2 .

Recall:

Theorem 7.3. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Question. What should we use as our pivotal quantity?

Example 14.

Using the Pivotal Method, find a $100(1 - \alpha)\%$ CI for σ^2 .

Fact. Suppose $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, a $100(1 - \alpha)\%$ confidence interval for σ^2 is

Example 15. The ages of a random sample of six NBA players are 28, 26, 34, 39, 29, and 29. Using this information, find a 95% CI for the population variance of the ages of all NBA players, assuming that the ages of NBA players are normally distributed.